Bootcamp - Git(Hub) : versionner & collaborer (2/2)

Xavier Clerc - xavier.clerc@enpc.fr
6 novembre 2026

Plan

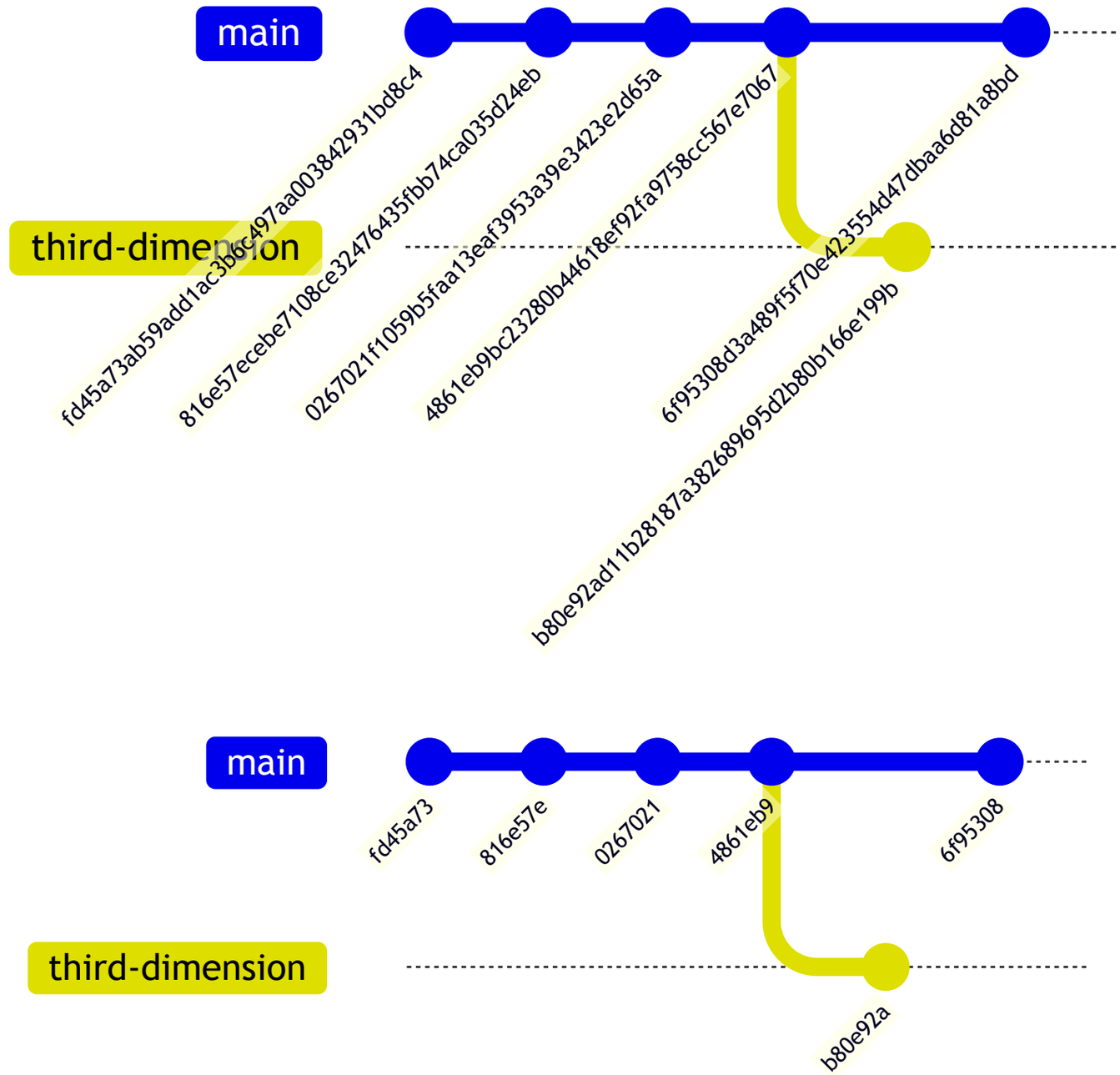
- Branches
- Distribution
- Manipulations
- Pour aller plus loin

Branches

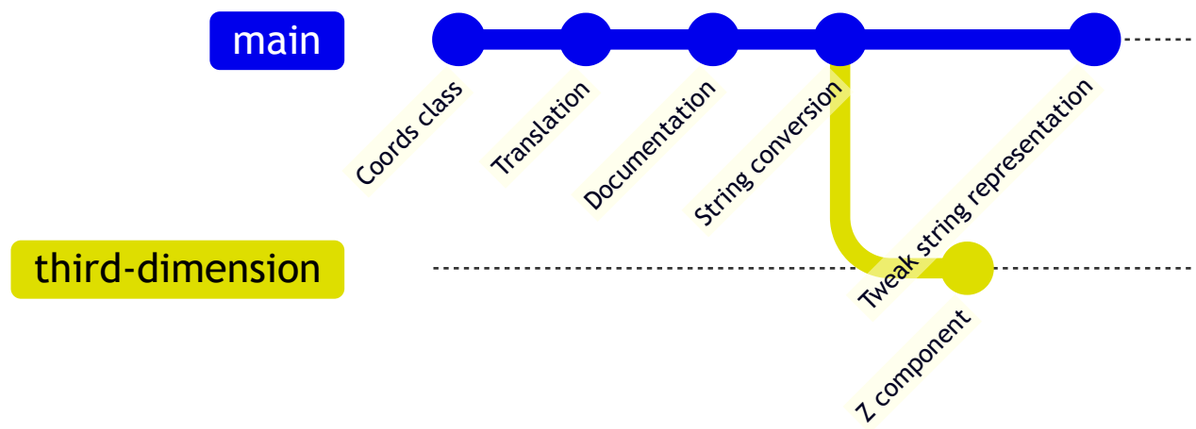
Branches

- Permettre un historique non-linéaire...
- ... pour faciliter les expérimentations
- ... pour simplifier la collaboration

Historique


main
third-dimension
fd45a73ab59add1ac3b6c497aa00384931bd8c4
816e57ecebe7108ce32476435fbb74ca035d24eb
0267021f1059b5faa13eaf3953a39e3423e2d65a
4861eb9bc23280b44618ef92fa9758cc567e7067
6f95308d3a489f5f70e423554d47dbaa6d81a8bd
b80e92ad11b28187a382689695d2b80b166e199b
main
third-dimension
fd45a73
816e57e
0267021
4861eb9
6f95308
b80e92a

Historique

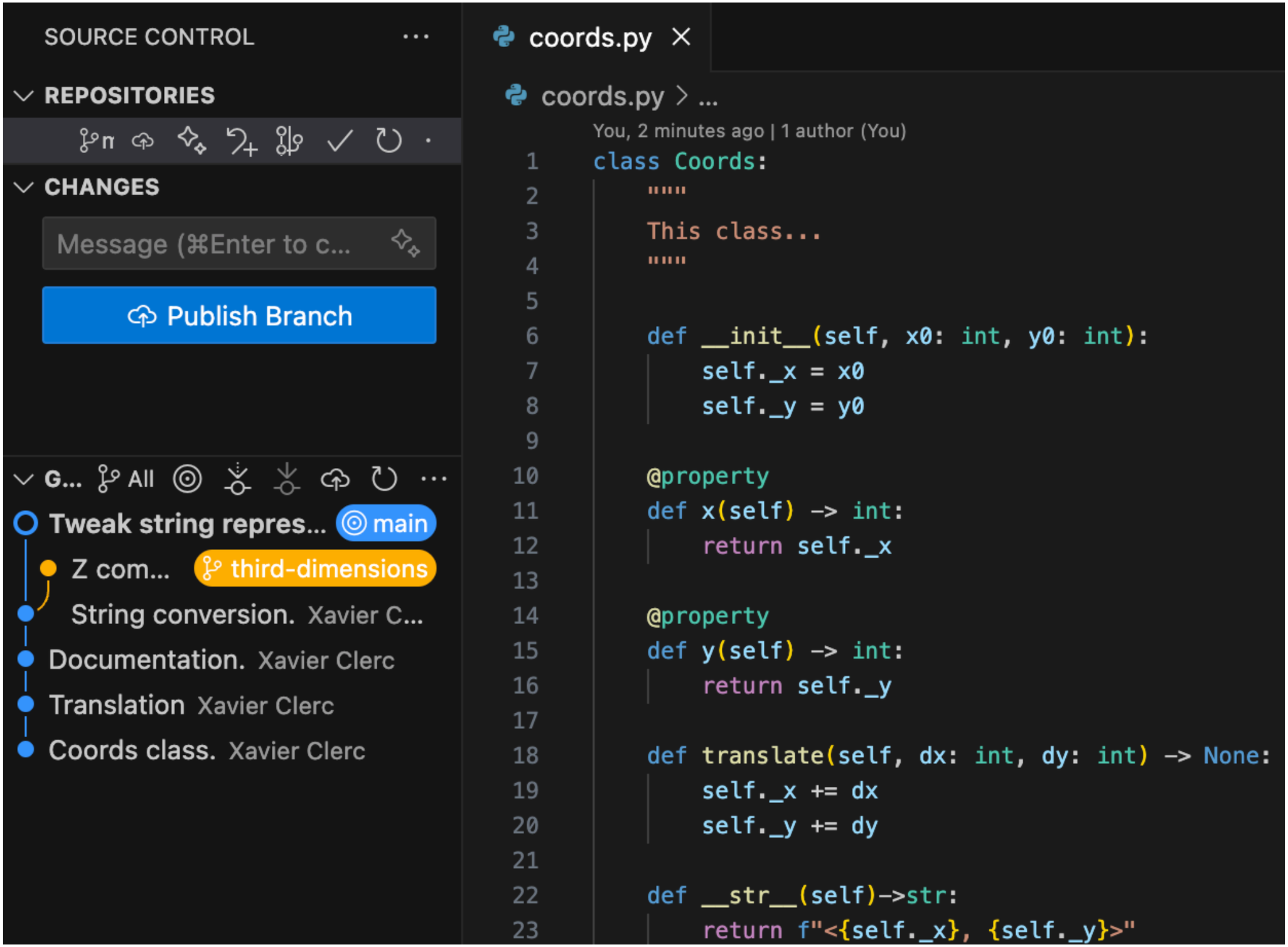

main
Coords class
Translation
Documentation
String conversion
Tweak string representation
third-dimension
Z component

Branches

- `git branch`

- `git branch new-branch-name`

- `git switch branch-name`


SOURCE CONTROL
REPOSITORIES
CHANGES
Message (⌘Enter to c...
Publish Branch
G...
All
Tweak string repres...
main
Z com...
third-dimensions
String conversion. Xavier C...
Documentation. Xavier Clerc
Translation Xavier Clerc
Coords class. Xavier Clerc
coords.py
coords.py > ...
You, 2 minutes ago | 1 author (You)
class Coords:
"""
This class...
"""
def __init__(self, x0: int, y0: int):
self._x = x0
self._y = y0
@property
def x(self) -> int:
return self._x
@property
def y(self) -> int:
return self._y
def translate(self, dx: int, dy: int) -> None:
self._x += dx
self._y += dy
def __str__(self)->str:
return f"<{self._x}, {self._y}>"

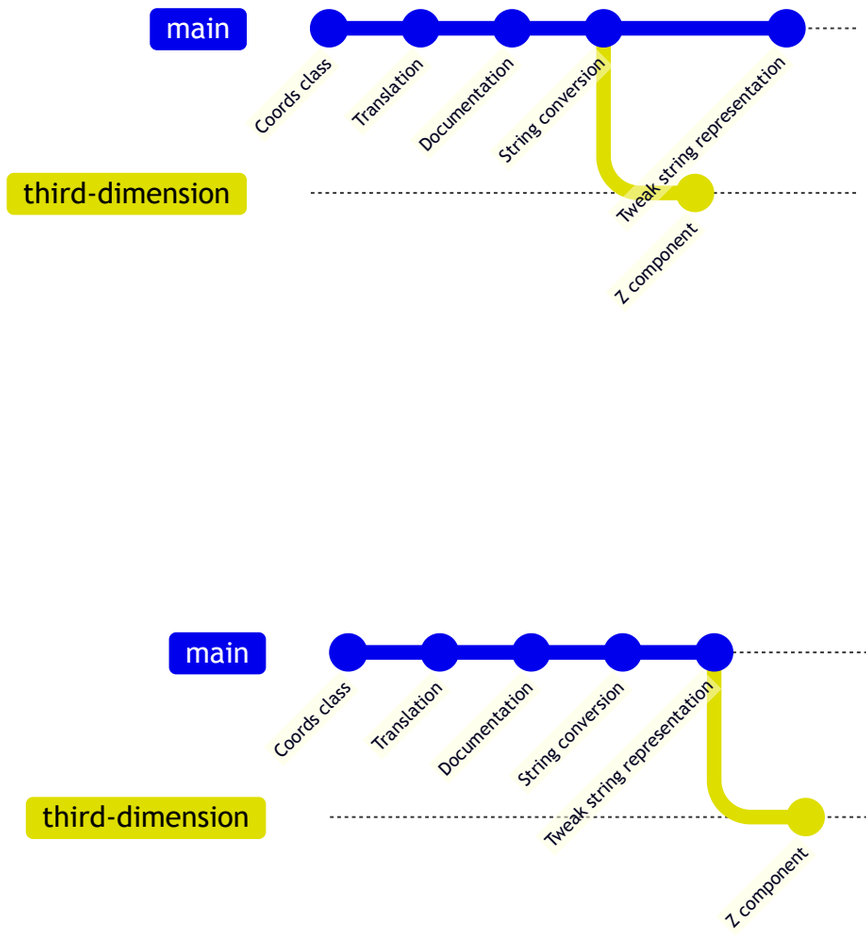
Rebase

- Travail sur la branche auxiliaire...

- ... pas terminé

- ... besoin des changements sur la branche principale

Rebase


main
Coords class
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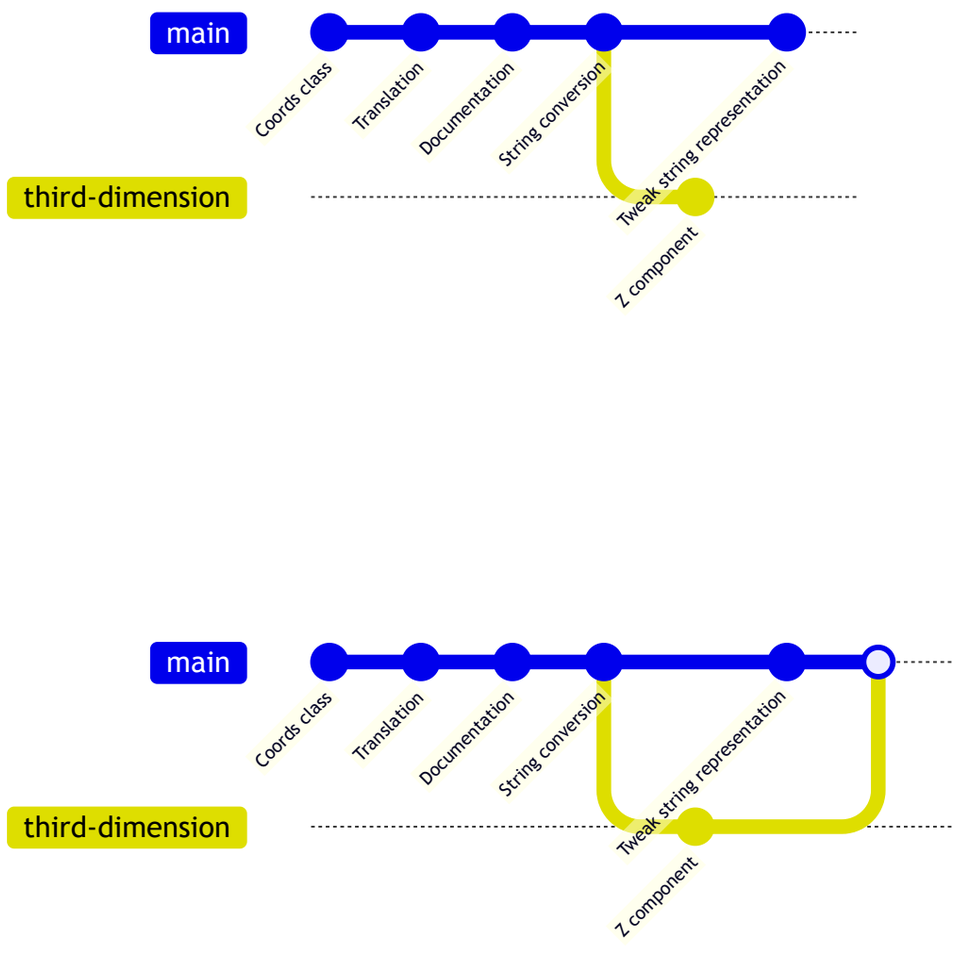
Merge

- Travail sur la branche auxiliaire...

- ... terminé

- ... application des changements à la branche principale

Merge


main
Coords class
Translation
Documentation
String conversion
Tweak string representation
third-dimension
Z component

Conflits

- Git garde la trace des lignes modifiées...
- ... pour détecter des modifications concurrentes
- ... et demander à l'utilisateur d'arbitrer
- Facilite grandement le travail à plusieurs
- (Critère purement textuel)

Conflits

```
class Coords:

    ...

    def translate(self, dx: int, dy: int, dz: int) -> None:
        self._x += dx
        self._y += dy
        self._z += dz

    def __str__(self)->str:
<<<<<<< HEAD
        return f"<{self._x}, {self._y}>"
=======
        return f"({self._x}, {self._y}, {self._z})"
>>>>>>> b80e92a (Z component.)
```

Conflits

- Marqueurs pour identifier les versions en jeu

- <<<<<<<< début première version

- ======== séparateur des versions

- >>>>>>>> fin deuxième version

Résolution

- `git rebase other-branch`

- Éditer le(s) fichier(s) pour enlever les marqueurs

- `git status`

- `git add ...`

Résolution

- Progrès commit par commit

- `git rebase --continue`

- (Commits sans conflits traités automatiquement)

- Abandon par `git rebase --abort`

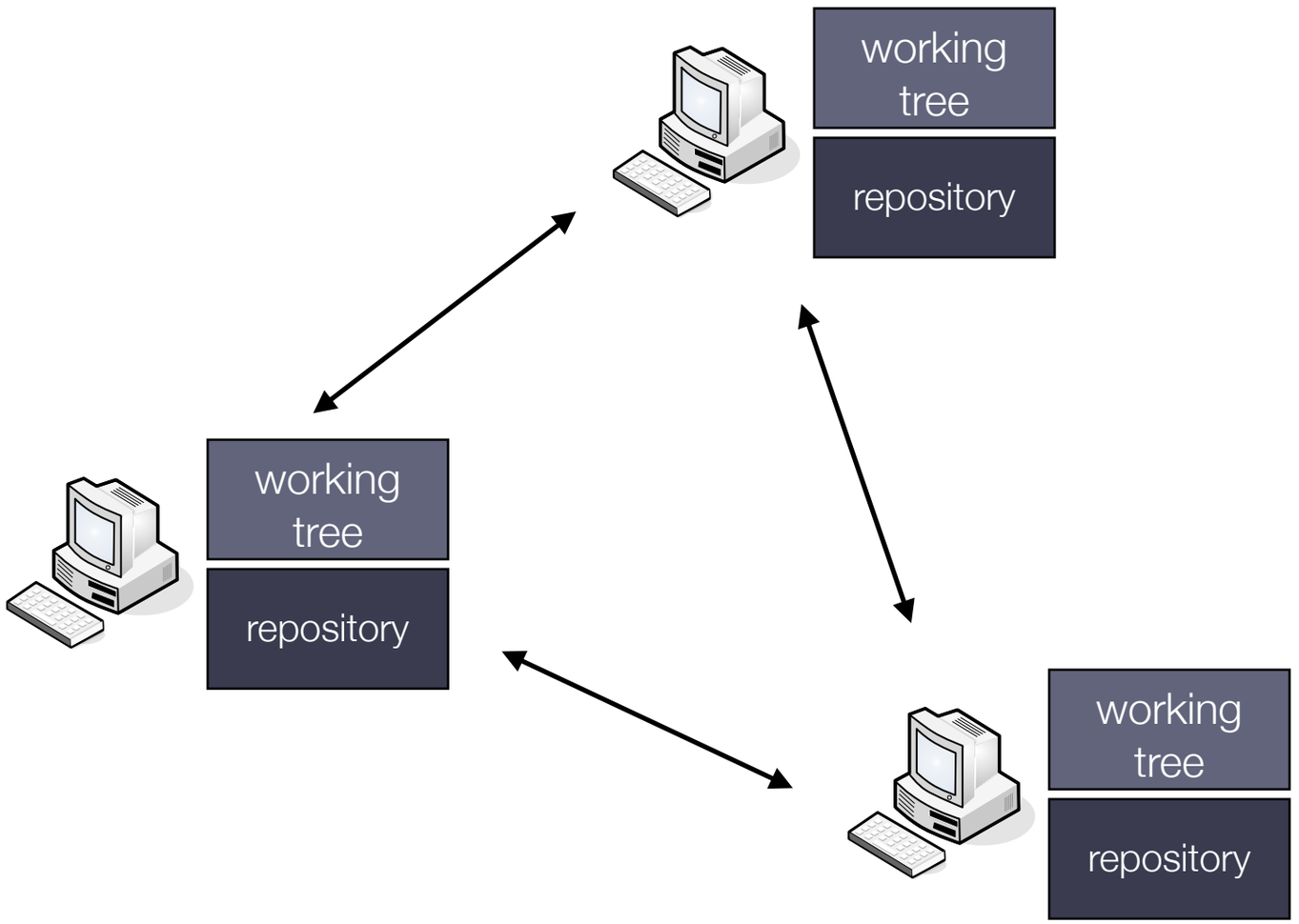
Interface

```
def compute_frequencies(text: str, ignore: set[str]) -> dict[str, float]:
Accept Current | Ignore
    occurrences: dict[str, int] = {}
    total = 0
    for word in re.findall(r"\b\w+\b", text.lower()):
        if word not in ignore:
            total += 1
            occurrences[word] = occurrences[word] + 1
    return {word: (occs / total) for word, occs in occurrences.items()}
```

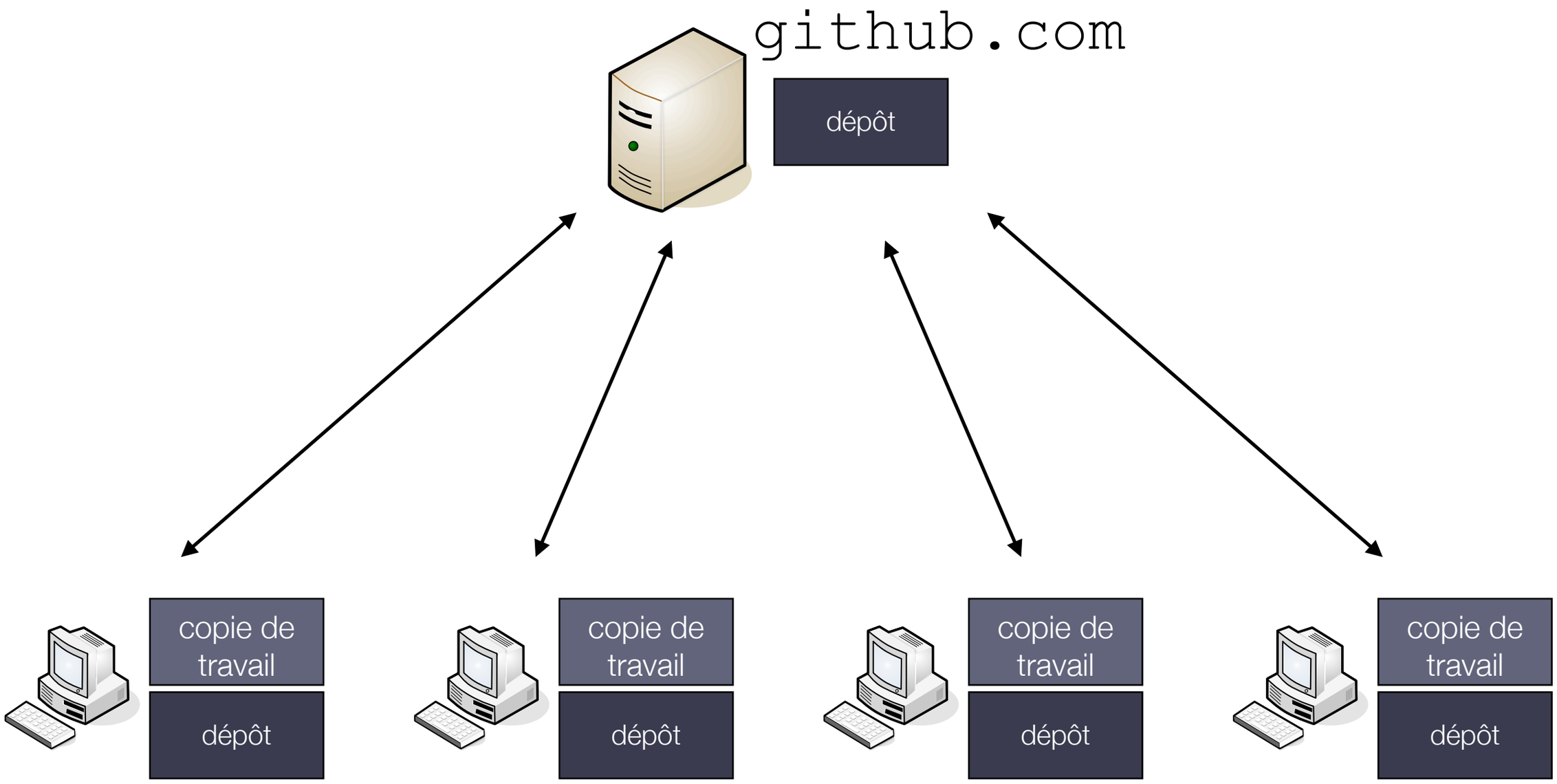
```
def compute_frequencies(text: str, ignore: set[str]) -> dict[str, float]:
Accept Incoming | Ignore
    """
    Compute the word frequencies in the passed text, ignoring the words
    present in the `ignore` set, and converting the words to lower case.
    """
    occurrences: dict[str, int] = {}
    total = 0
    for word in re.findall(r"\b\w+\b", text.lower()):
        if word not in ignore:
            total += 1
            occurrences[word] = occurrences.get(word, 0) + 1
    return {word: (occs / total) for word, occs in occurrences.items()}
```

Distribution

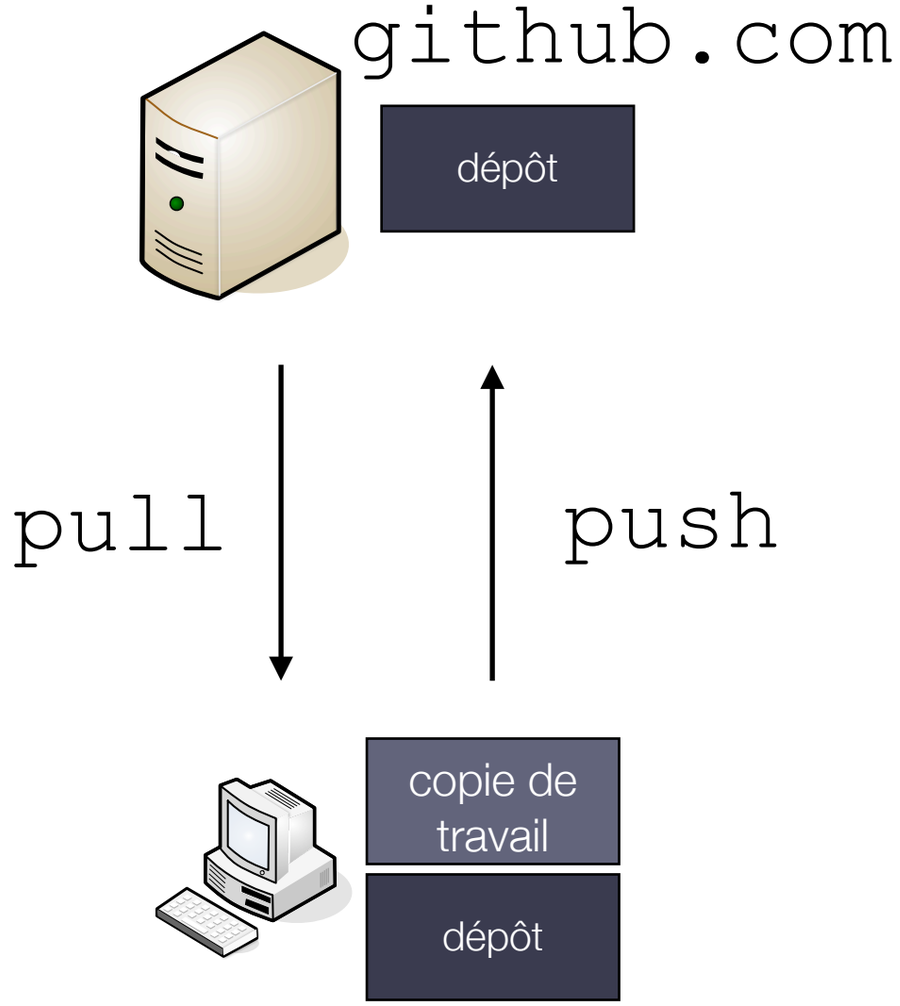
Théorie


working
tree
repository
working
tree
repository
working
tree
repository

Pratique


github.com
dépôt
copie de travail
dépôt
copie de travail
dépôt
copie de travail
dépôt
copie de travail
dépôt

Opérations distantes



Pull request

- *Request to pull code from another branch to merge it*
- Revue de code (*code review*)
- Discussion sur le changement proposé...
- ... correction
- ... qualité
- Sera utilisé en 1PROG pour rendre les exercices

Pull request

- Les changements dans VS Code sont éphémères

- Ils doivent être poussés vers le *fork*

- La *pull request* est alors créée depuis le *fork*

xclerc-testing/2425-TP2-Draft | cellular_automaton.py — 2425-

https://super-sniffle-v6qww7x5xgg9cwxr6.github.dev

2425-TP2-Draft [Codespaces: super sniffle]

SOURCE CONTROL

SOURCE CONTROL

message

✓ Commit

Changes 1

cellular_automaton.py M

Stage Changes

SOURCE CONTROL GRAPH Auto

Fix CI. Xavier Clerc main

Fix CI. Xavier Clerc

Fix CI. Xavier Clerc

Fix CI. Xavier Clerc

Fix CI. Xavier Clerc

Add CI. Xavier Clerc

Add pull request template. Xavier Clerc

Add ".gitconfig" file. Xavier Clerc

Add Live Share extension. Xavier Clerc

Missing typing annotation. Xavier Clerc

Apply black. Xavier Clerc

Add source. Xavier Clerc

Minor changes. Xavier Clerc

Draft Xavier Clerc

[Preview] README.md

cellular_automaton.py M

cellular_automaton.py > ...

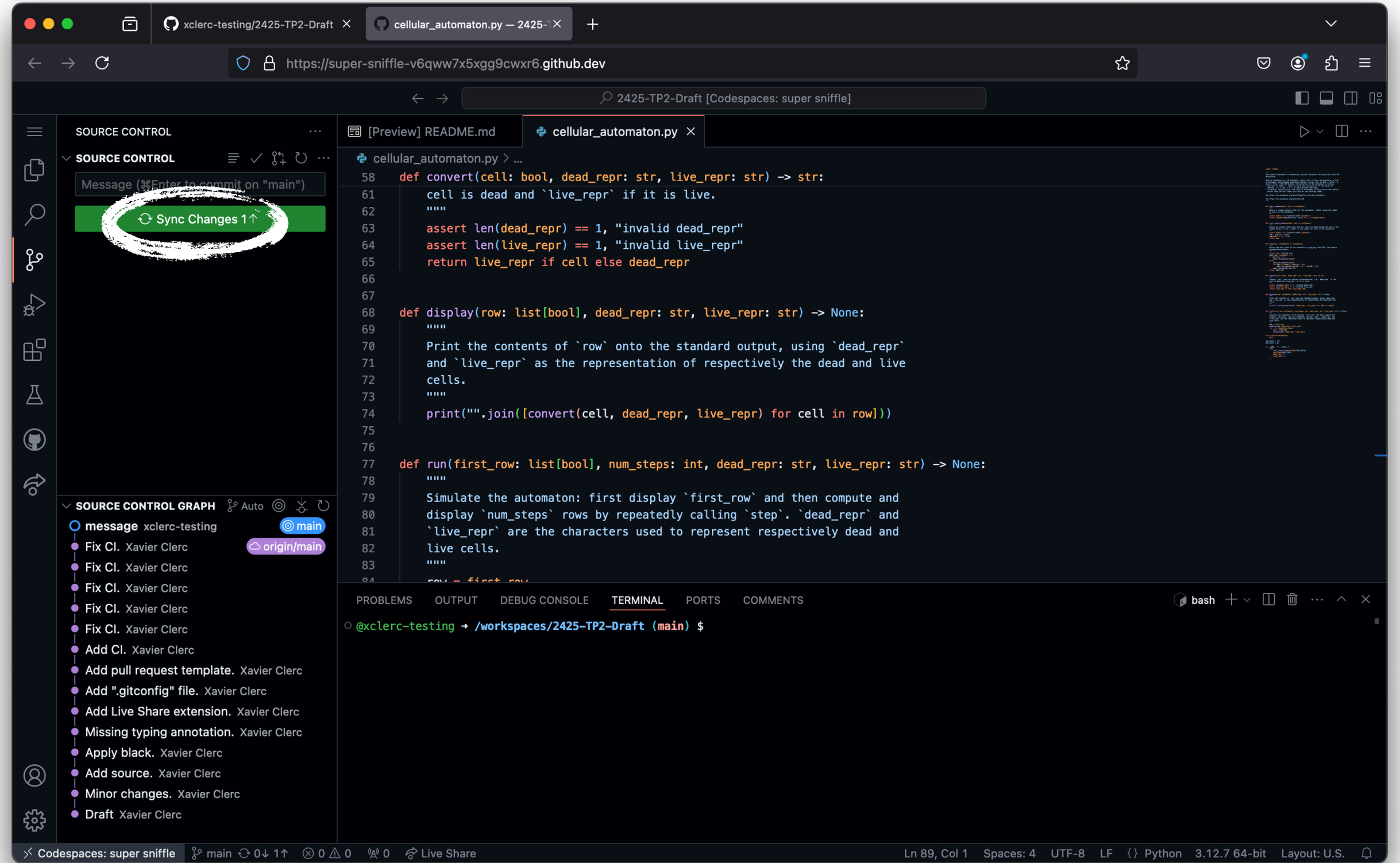
```
58  def convert(cell: bool, dead_repr: str, live_repr: str) -> str:
61      cell is dead and `live_repr` if it is live.
62      """
63      assert len(dead_repr) == 1, "invalid dead_repr"
64      assert len(live_repr) == 1, "invalid live_repr"
65      return live_repr if cell else dead_repr
66
67
68  def display(row: list[bool], dead_repr: str, live_repr: str) -> None:
69      """
70      Print the contents of `row` onto the standard output, using `dead_repr`
71      and `live_repr` as the representation of respectively the dead and live
72      cells.
73      """
74      print("".join([convert(cell, dead_repr, live_repr) for cell in row]))
75
76
77  def run(first_row: list[bool], num_steps: int, dead_repr: str, live_repr: str) -> None:
78      """
79      Simulate the automaton: first display `first_row` and then compute and
80      display `num_steps` rows by repeatedly calling `step`. `dead_repr` and
81      `live_repr` are the characters used to represent respectively dead and
82      live cells.
83      """
```

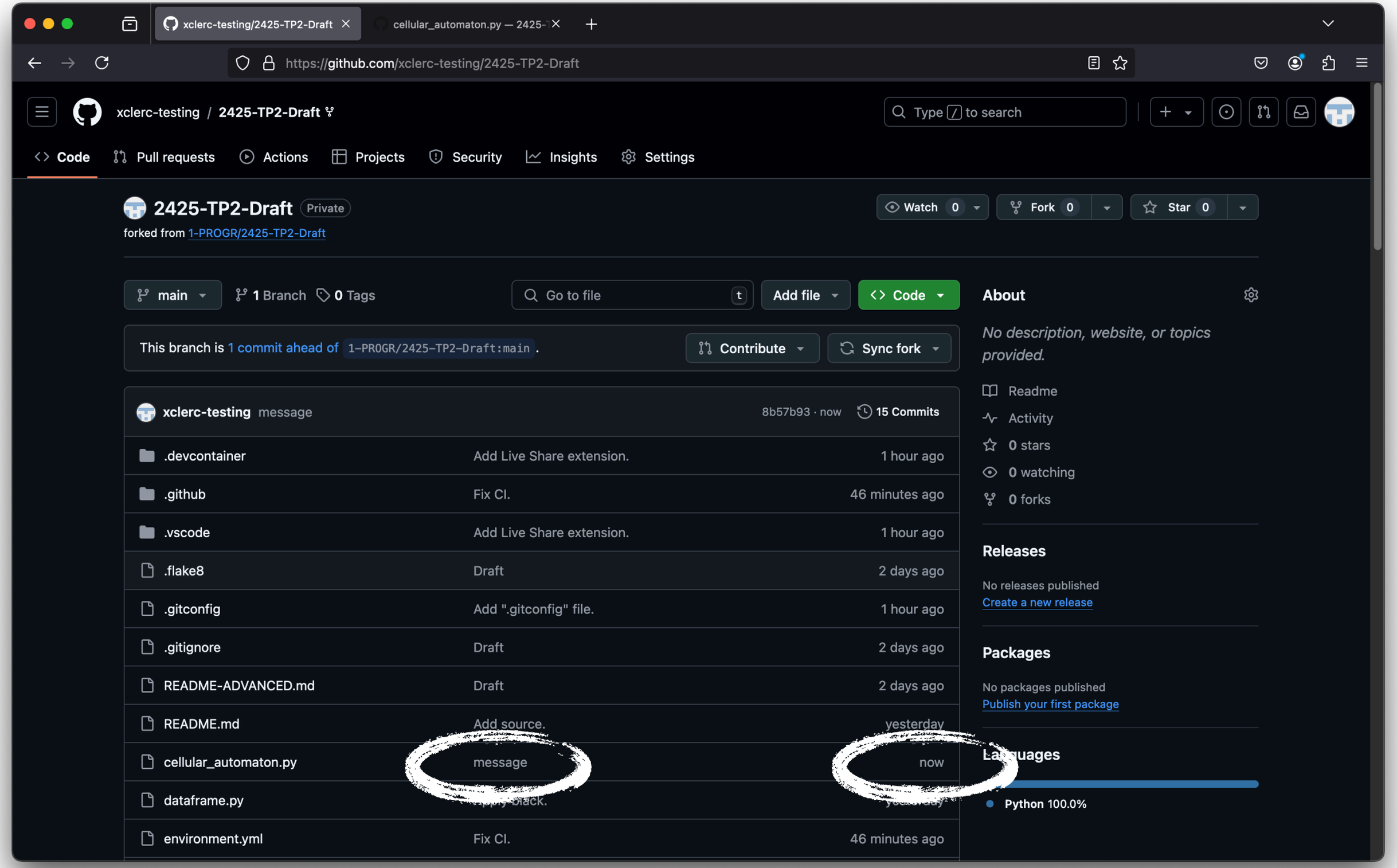
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

bash

```
@xclerc-testing → /workspaces/2425-TP2-Draft (main) $
```

Codespaces: super sniffle main* 0 0 0 Live Share Ln 89, Col 1 Spaces: 4 UTF-8 LF Python 3.12.7 64-bit Layout: U.S.


xclerc-testing/2425-TP2-Draft
cellular_automaton.py — 2425-
https://super-sniffle-v6qww7x5xgg9cwxr6.github.dev
2425-TP2-Draft [Codespaces: super sniffle]
SOURCE CONTROL
SOURCE CONTROL
Message (⌘Enter to commit on "main")
Sync Changes 1↑
SOURCE CONTROL GRAPH
Auto
message xclerc-testing
main
Fix CI. Xavier Clerc
origin/main
Fix CI. Xavier Clerc
Fix CI. Xavier Clerc
Fix CI. Xavier Clerc
Fix CI. Xavier Clerc
Add CI. Xavier Clerc
Add pull request template. Xavier Clerc
Add ".gitconfig" file. Xavier Clerc
Add Live Share extension. Xavier Clerc
Missing typing annotation. Xavier Clerc
Apply black. Xavier Clerc
Add source. Xavier Clerc
Minor changes. Xavier Clerc
Draft Xavier Clerc
[Preview] README.md
cellular_automaton.py
cellular_automaton.py > ...
58 def convert(cell: bool, dead_repr: str, live_repr: str) -> str:
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67
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83 """
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
COMMENTS
bash
@xclerc-testing → /workspaces/2425-TP2-Draft (main) $
Codespaces: super sniffle
main
0↓ 1↑
0 0 0
Live Share
Ln 89, Col 1
Spaces: 4
UTF-8
LF
Python
3.12.7 64-bit
Layout: U.S.


xclerc-testing/2425-TP2-Draft
cellular_automaton.py — 2425-
https://github.com/xclerc-testing/2425-TP2-Draft
xclerc-testing / 2425-TP2-Draft
Type / to search
Code
Pull requests
Actions
Projects
Security
Insights
Settings
2425-TP2-Draft
Private
forked from 1-PROGR/2425-TP2-Draft
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
This branch is 1 commit ahead of 1-PROGR/2425-TP2-Draft:main.
Contribute
Sync fork
xclerc-testing message
8b57b93 · now
15 Commits
.devcontainer Add Live Share extension. 1 hour ago
.github Fix CI. 46 minutes ago
.vscode Add Live Share extension. 1 hour ago
.flake8 Draft 2 days ago
.gitconfig Add ".gitconfig" file. 1 hour ago
.gitignore Draft 2 days ago
README-ADVANCED.md Draft 2 days ago
README.md Add source. yesterday
cellular_automaton.py message now
dataframe.py
environment.yml Fix CI. 46 minutes ago
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Languages
Python 100.0%

xclerc-testing/2425-TP2-Draft | cellular_automaton.py — 2425-

https://github.com/xclerc-testing/2425-TP2-Draft

xclerc-testing / 2425-TP2-Draft

Type / to search

Code | Pull requests | Actions | Projects | Security | Insights | Settings

2425-TP2-Draft

Private

forked from 1-PROGR/2425-TP2-Draft

Watch 0 | Fork 0 | Star 0

main | 1 Branch | 0 Tags | Go to file t | Add file | Code

This branch is 1 commit ahead of 1-PROGR/2425-TP2-Draft:main. | Contribute | Sync fork

xclerc-testing message — 8b57b93 · now — 15 Commits

.devcontainer	Add Live Share extension.	1 hour ago
.github	Fix CI.	46 minutes ago
.vscode	Add Live Share extension.	1 hour ago
.flake8	Draft	2 days ago
.gitconfig	Add ".gitconfig" file.	1 hour ago
.gitignore	Draft	2 days ago
README-ADVANCED.md	Draft	2 days ago
README.md	Add source.	yesterday
cellular_automaton.py	message	now
dataframe.py	Apply black.	yesterday
environment.yml	Fix CI.	46 minutes ago

About

No description, website, or topics provided.

Readme
Activity
0 stars
0 watching
0 forks

Releases

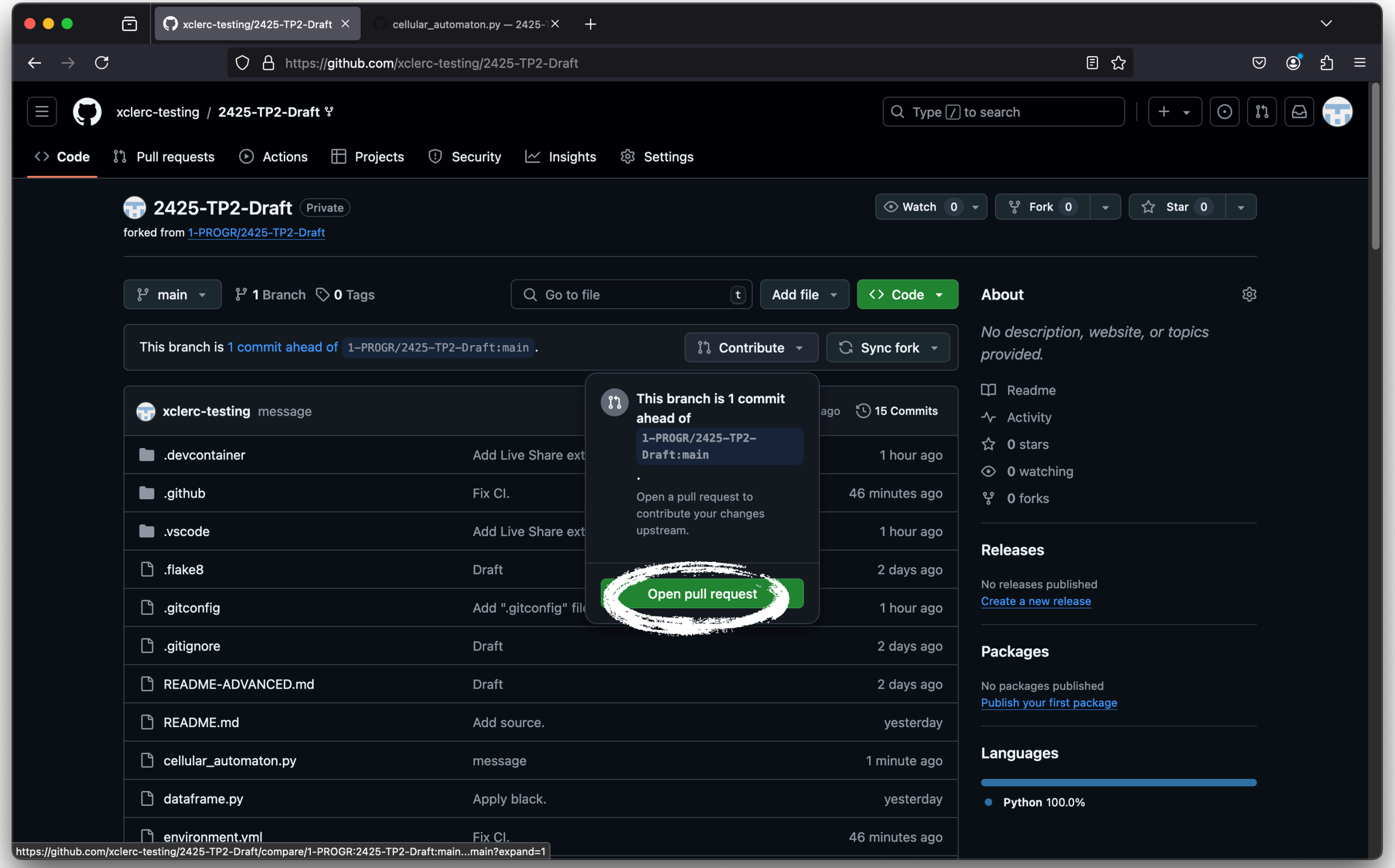
No releases published
Create a new release

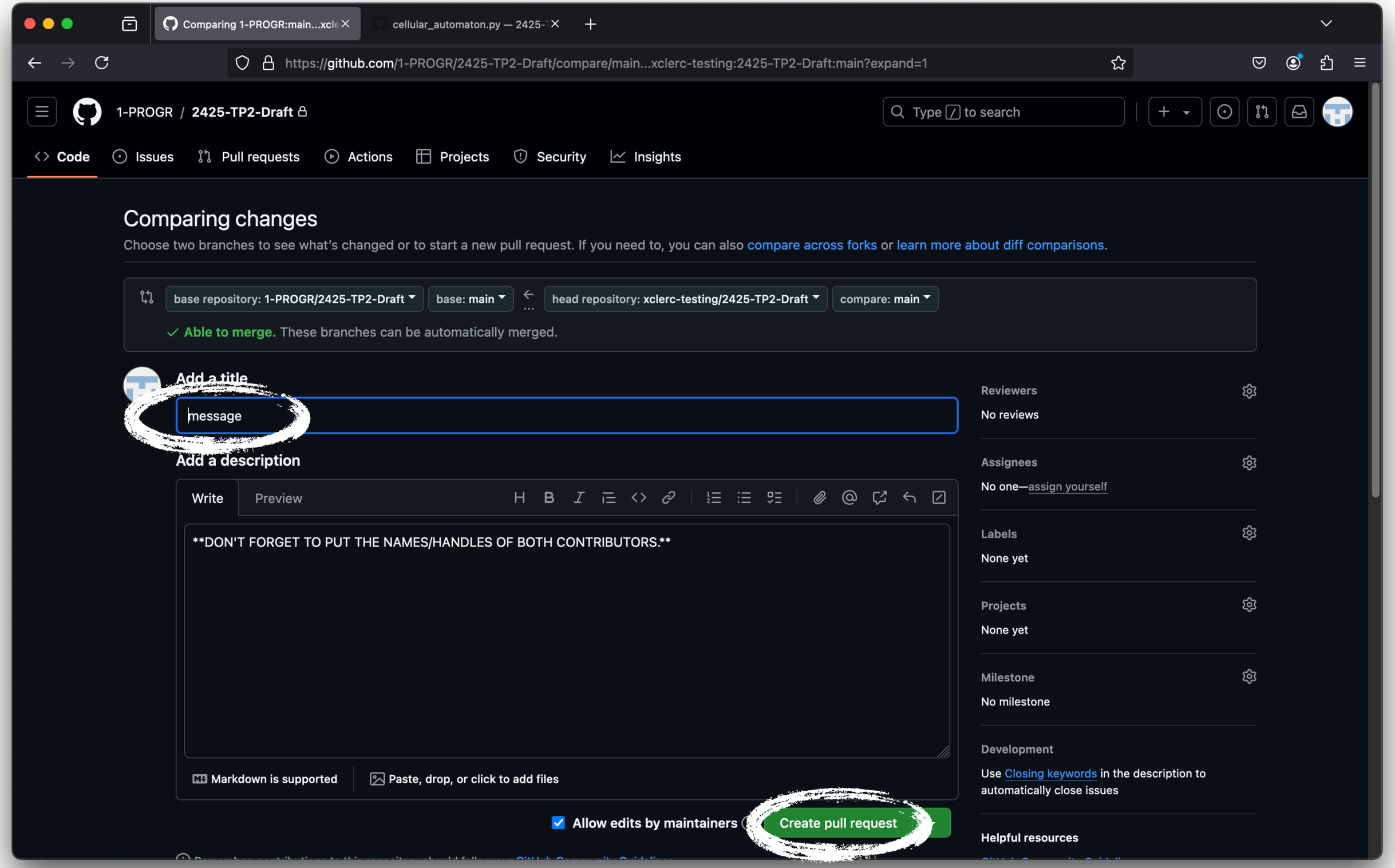
Packages

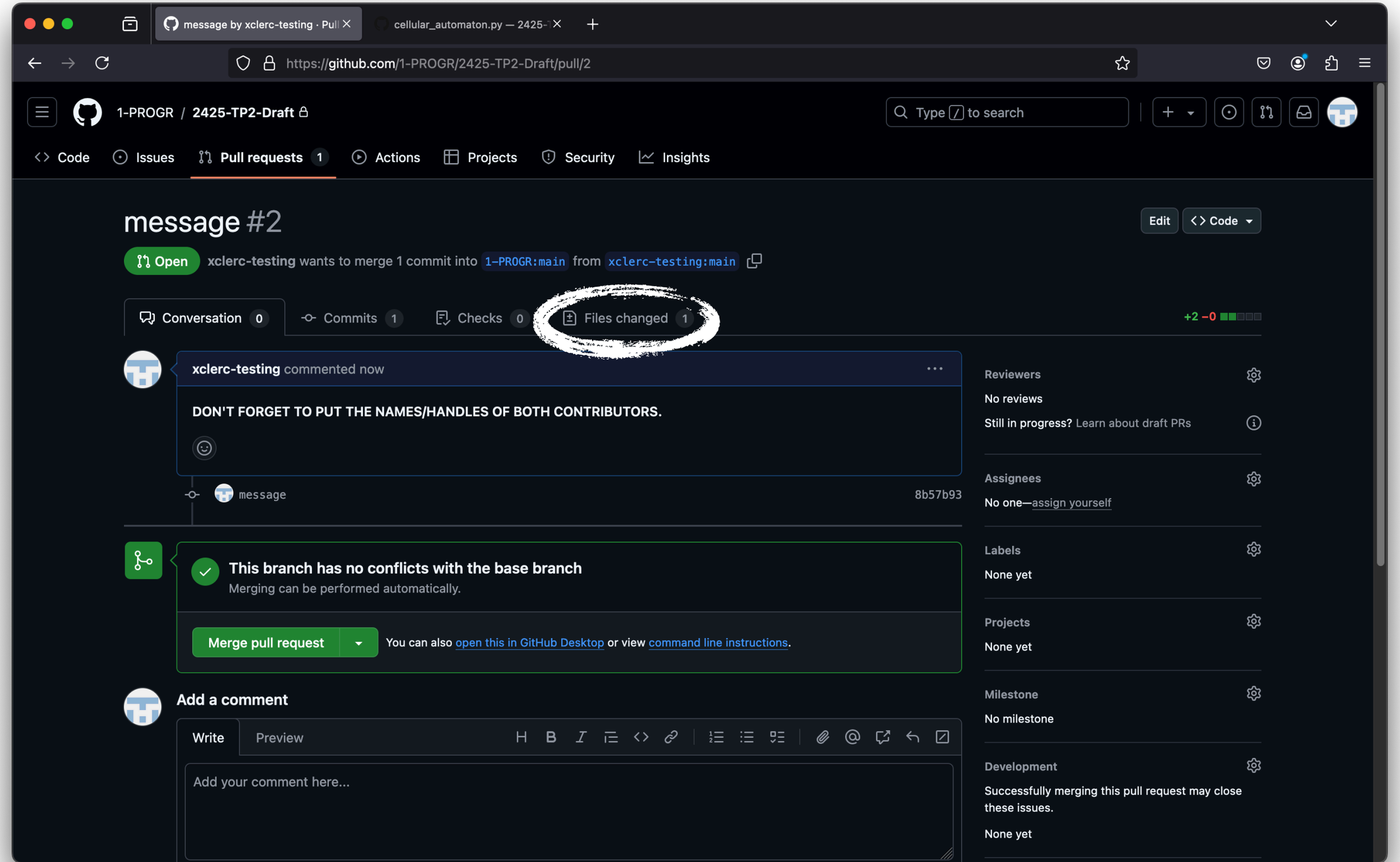
No packages published
Publish your first package

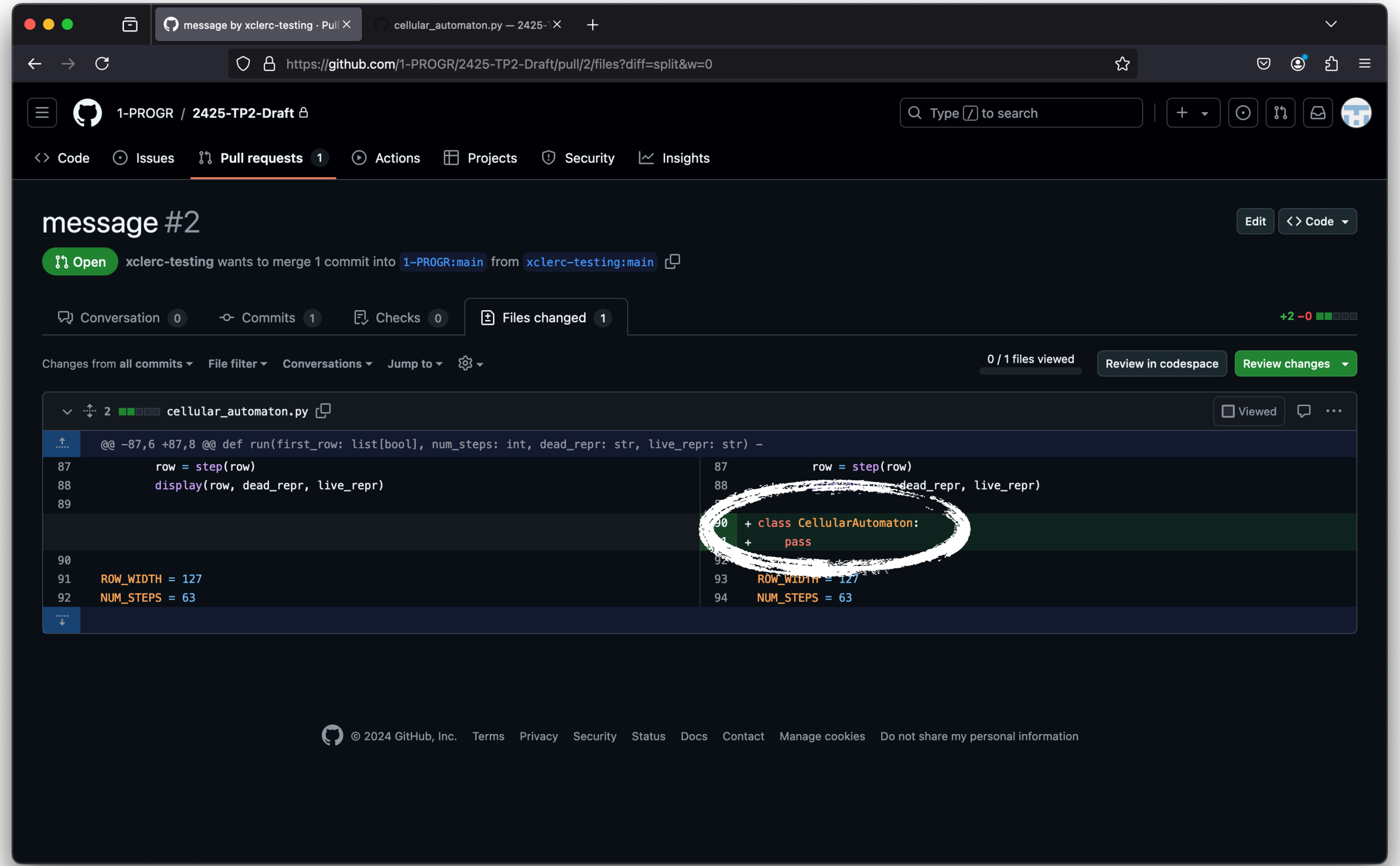
Languages

Python 100.0%

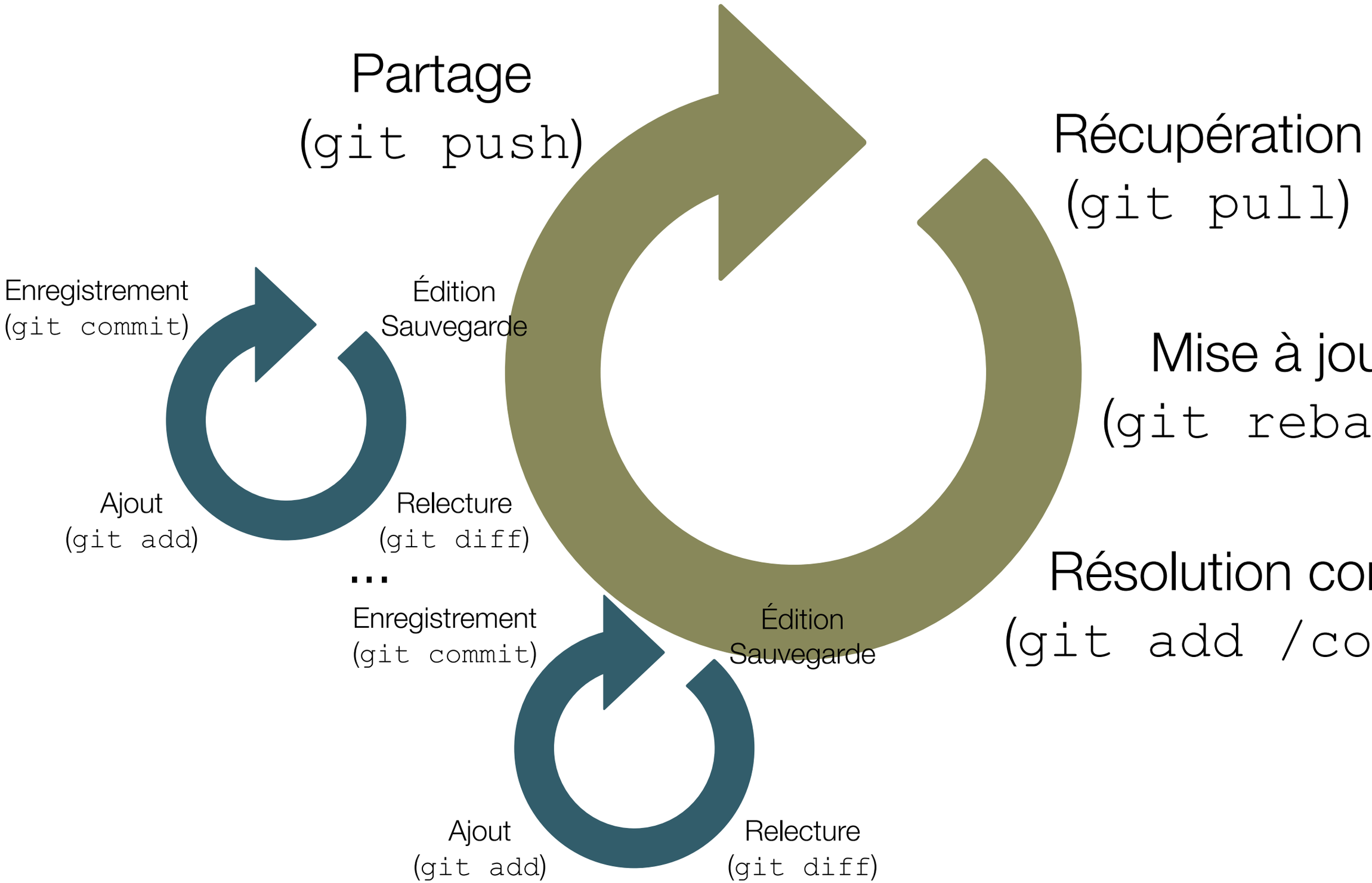

xclerc-testing/2425-TP2-Draft
cellular_automaton.py — 2425-
https://github.com/xclerc-testing/2425-TP2-Draft
xclerc-testing / 2425-TP2-Draft
Type / to search
Code
Pull requests
Actions
Projects
Security
Insights
Settings
2425-TP2-Draft
Private
forked from 1-PROGR/2425-TP2-Draft
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
t
Add file
Code
This branch is 1 commit ahead of 1-PROGR/2425-TP2-Draft:main .
Contribute
Sync fork
This branch is 1 commit ahead of 1-PROGR/2425-TP2-Draft:main .
Open a pull request to contribute your changes upstream.
Open pull request
xclerc-testing message
15 Commits
.devcontainer Add Live Share ext 1 hour ago
.github Fix CI. 46 minutes ago
.vscode Add Live Share ext 1 hour ago
.flake8 Draft 2 days ago
.gitconfig Add ".gitconfig" fil 1 hour ago
.gitignore Draft 2 days ago
README-ADVANCED.md Draft 2 days ago
README.md Add source. yesterday
cellular_automaton.py message 1 minute ago
dataframe.py Apply black. yesterday
environment.yml Fix CI. 46 minutes ago
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Languages
Python 100.0%
https://github.com/xclerc-testing/2425-TP2-Draft/compare/1-PROGR:2425-TP2-Draft:main...main?expand=1


Comparing 1-PROGR:main...xcle
cellular_automaton.py — 2425-
https://github.com/1-PROGR/2425-TP2-Draft/compare/main...xclerc-testing:2425-TP2-Draft:main?expand=1
1-PROGR / 2425-TP2-Draft
Type / to search
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Comparing changes
Choose two branches to see what's changed or to start a new pull request. If you need to, you can also compare across forks or learn more about diff comparisons.
base repository: 1-PROGR/2425-TP2-Draft
base: main
head repository: xclerc-testing/2425-TP2-Draft
compare: main
Able to merge. These branches can be automatically merged.
Add a title
message
Add a description
Write
Preview
DON'T FORGET TO PUT THE NAMES/HANDLES OF BOTH CONTRIBUTORS.
Markdown is supported
Paste, drop, or click to add files
Allow edits by maintainers
Create pull request
Reviewers
No reviews
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Use Closing keywords in the description to automatically close issues
Helpful resources


message by xclerc-testing · Pull
cellular_automaton.py — 2425-
https://github.com/1-PROGR/2425-TP2-Draft/pull/2
1-PROGR / 2425-TP2-Draft
Type / to search
Code
Issues
Pull requests 1
Actions
Projects
Security
Insights
message #2
Edit
Code
Open
xclerc-testing wants to merge 1 commit into 1-PROGR:main from xclerc-testing:main
Conversation 0
Commits 1
Checks 0
Files changed 1
+2 −0
xclerc-testing commented now
DON'T FORGET TO PUT THE NAMES/HANDLES OF BOTH CONTRIBUTORS.
message
8b57b93
This branch has no conflicts with the base branch
Merging can be performed automatically.
Merge pull request
You can also open this in GitHub Desktop or view command line instructions.
Add a comment
Write
Preview
Add your comment here...
Reviewers
No reviews
Still in progress? Learn about draft PRs
Assignees
No one—assign yourself
Labels
None yet
Projects
None yet
Milestone
No milestone
Development
Successfully merging this pull request may close these issues.
None yet


message by xclerc-testing · Pull
cellular_automaton.py — 2425-
https://github.com/1-PROGR/2425-TP2-Draft/pull/2/files?diff=split&w=0
1-PROGR / 2425-TP2-Draft
Type / to search
Code
Issues
Pull requests 1
Actions
Projects
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Insights
message #2
Edit
Code
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xclerc-testing wants to merge 1 commit into 1-PROGR:main from xclerc-testing:main
Conversation 0
Commits 1
Checks 0
Files changed 1
+2 −0
Changes from all commits
File filter
Conversations
Jump to
0 / 1 files viewed
Review in codespace
Review changes
2 cellular_automaton.py
Viewed
@@ -87,6 +87,8 @@ def run(first_row: list[bool], num_steps: int, dead_repr: str, live_repr: str) -
87 row = step(row)
88 display(row, dead_repr, live_repr)
89
90 + class CellularAutomaton:
91 + pass
90
91 ROW_WIDTH = 127
92 NUM_STEPS = 63
87 row = step(row)
88 dead_repr, live_repr)
93 ROW_WIDTH = 127
94 NUM_STEPS = 63
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Security
Status
Docs
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Cycle de travail



Mise à jour
(`git rebase`)

Résolution conflits
(`git add /commit`)

Manipulations

Manipulations

- VS Code & GitHub

- Utilisation de branches présentées

- Résolution de conflits

Pour aller plus loin

Pour aller plus loin

- Documentation :
- https://git-scm.com/docs
- Aide-mémoire :
- https://education.github.com/git-cheat-sheet-education.pdf
- Simulation :
- https://git-school.github.io/visualizing-git/